ChatGPT

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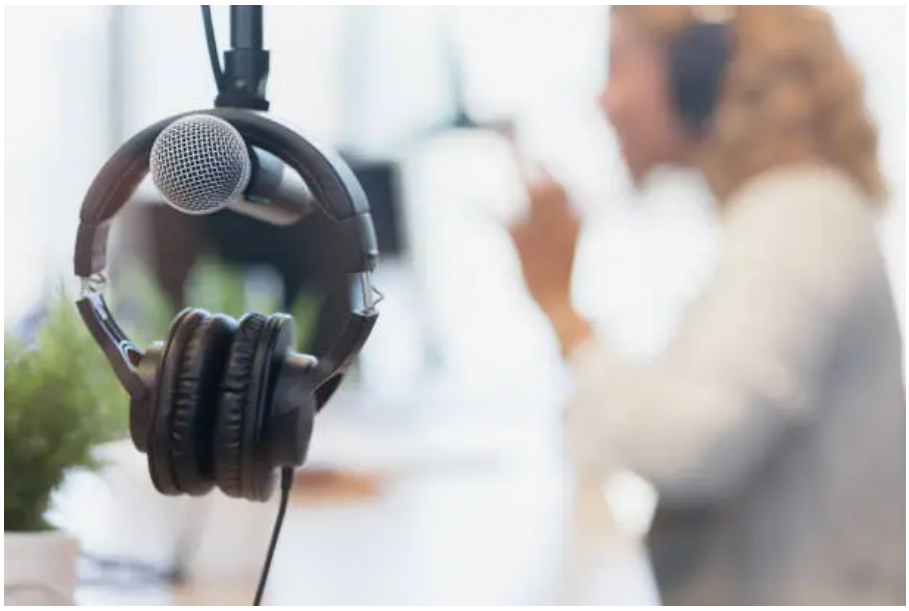
EchoHub - Project Documentation

EchoHub is an interactive platform that merges podcast browsing with AI-powered content creation. Users can explore a curated collection of podcast episodes by categories or tags, play audio seamlessly in a built-in mini-player, and even generate brand-new episode scripts using the site's integrated AI studio. Each episode page displays a detailed description, full transcript, and any related media (such as embedded video). The homepage uses large cover illustrations and modern typography to engage the user.



Website Structure

- **Home:** Displays featured categories or episodes with cover images and titles.
- **Category Pages:** Show a list of episodes within a topic area (e.g., Technology, Science), each as a clickable card.
- **Episode Detail:** Presents the episode's title, author, description, transcript, and playback controls (audio and video if available).
- **AI Studio:** A special page where users input a topic or title to generate a complete podcast script with one button click.
- **About:** Provides project overview and team information (including an embedded "About Us" audio introduction).
- **Persistent Player:** A mini-player docked at the bottom of every page, so playback continues uninterrupted as users browse.

The site's layout follows a clear navigation flow. The top header includes the EchoHub logo and search bar, and a sidebar or menu lists the categories. All pages are linked smoothly via this navigation and the footer player. The persistent player ensures that audio never stops when navigating between pages

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Key Features

EchoHub offers a range of user-focused features designed for seamless interaction:

- **Real-time Search & Filtering:** Users can search episodes by keywords or click on displayed tags to quickly find relevant content.
- **Content-rich Episode Pages:** Every episode includes cover art, title, description, creator name, and a complete text transcript for accessibility.
- **Interactive Audio Player:** The sticky bottom player provides play/pause, a seek bar, and volume controls. It displays the current speed (1x) and maintains position even when the user switches pages.
- **Favorites & History:** Allows marking episodes as favorites, and the app records recently played items for easy access.
- **Light/Dark Mode:** A toggle lets users switch between light and dark themes for comfortable viewing in different lighting conditions.
- **AI Script Writer:** On the AI Studio page, users enter a topic and the system generates a podcast script automatically. The script is then read aloud using text-to-speech in real time.

These features emphasize both usability and creativity, blending straightforward navigation with powerful content tools

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User Interface (UI) and Flow

The UI uses a clean, modern design with a pastel color palette. Key components include:

- **Header:** Displays the brand name ("EchoHub") and a search input. Category filters or navigation links are also present.
- **Episode Cards:** Each card shows the episode's cover image and title. Clicking a card expands to the detail view or toggles playback.
- **Player Controls:** The mini-player (visible on all pages) shows episode info and allows control of playback.
- **Episode Detail Layout:** Organized content area with title at top, a 'Play' button, episode details below, followed by the transcript text.
- **AI Studio Interface:** A simple text editor field and a "Generate Script" button. Once generated, a "Play Voice" button appears to listen to the AI-created script.
- **Navigation Flow:** Users typically start on the home page, browse categories or use search, click an episode to open its detail, and then use the player. The consistent layout and the persistent player make this flow smooth and intuitive.

EchoHub's design focuses on clarity and consistency. The intuitive layout ensures that users always know where they are and how to control playback

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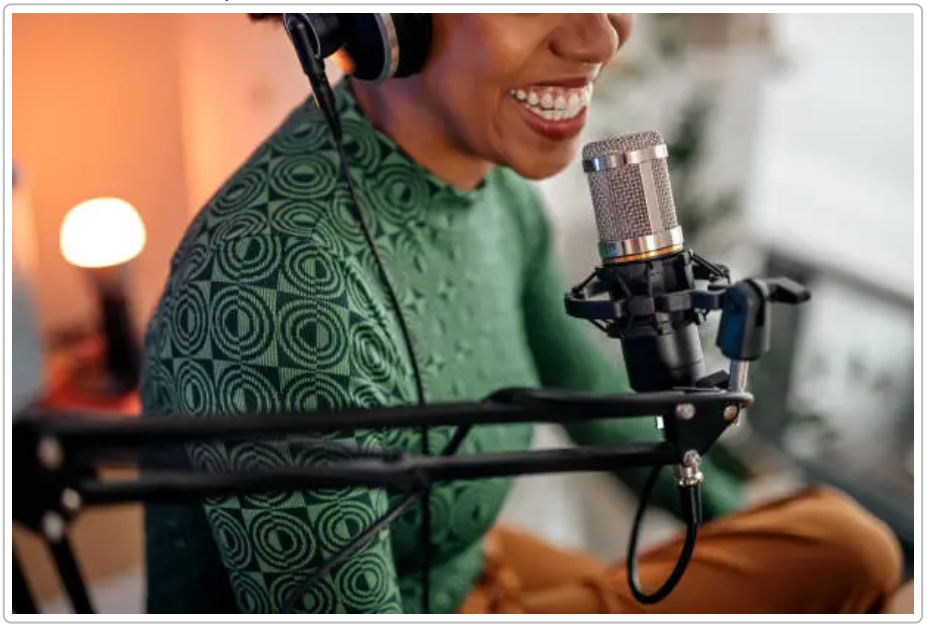
Team & Roles

The EchoHub project was built by a team of five students, each contributing key skills. We collaborated on design, content, and development from start to finish



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- **Nikhil:** Lead developer. Responsible for the website structure, front-end coding (HTML/CSS/JavaScript), and integrating the audio player and search functionality.
- **Bhumika:** Content creator. Researched podcast topics, wrote initial scripts, and recorded and edited audio segments.
- **Maleeha:** Audio & design. Recorded and mixed podcast audio, and crafted custom cover illustrations for episodes.

- **Ria:** Scriptwriter and researcher. Developed episode scripts (using the AI tools), verified factual content, and managed the project documentation.
- **Jasmine:** Designer and coordinator. Created the slide decks and visuals for the project, helped set the overall aesthetic, and ensured visual consistency.

About Us Podcast

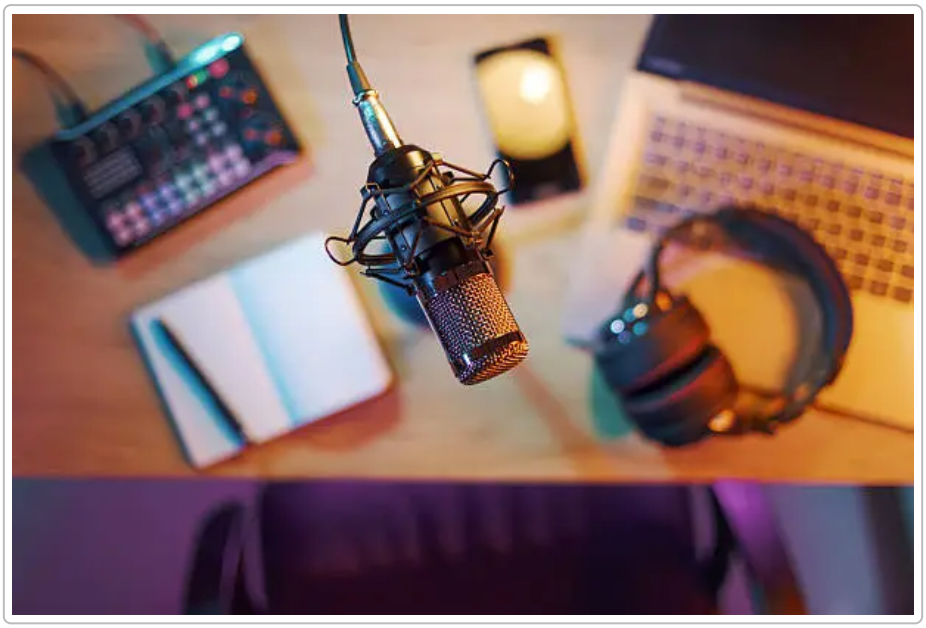
- **Title:** "About Us"
- **Category:** General
- **Duration:** 1:00 (one minute)
- **Description:** A brief introduction to the EchoHub platform and its creators.
- **Content:** In this episode, each team member introduces themselves and their role on the project (for example: "Hello, I'm Nikhil... Welcome to EchoHub, a creative podcast platform built by our team of five!").
- **Purpose:** The episode humanizes the project, allowing users to connect with the creators behind the content. The podcast is embedded on the About page and can be played via the persistent mini-player.

Technical Implementation

EchoHub is implemented as a static, client-side web application using standard web technologies:

- **Frontend:** Built with HTML5, CSS3, and JavaScript. Uses CSS Flexbox for responsive layout and CSS variables for managing light/dark theme colors.
- **Data & Content:** Episode metadata (titles, descriptions, tags) and transcripts are stored in a local JavaScript data file. Podcast audio files (`.mp3`) and cover images are included in the project's assets.
- **Audio Playback:** Utilizes the HTML5 `<audio>` element with JavaScript control for play/pause, seeking, and volume. The persistent player maintains playback state across pages.
- **Search & State:** Implements a dynamic search/filter by tags and text. Favorites and play history are saved using `localStorage` so preferences persist in the browser.
- **AI Script Studio:** Integrates a large language model via an API. When the user submits a topic, the model generates a script and the app then uses a text-to-speech engine to play it.
- **Hosting:** As a static site, EchoHub can be hosted on any web server or service (e.g., GitHub Pages) without requiring a backend.

The system's audio-centric design mirrors real-world podcast production equipment (e.g., microphone and mixer) as shown below



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Visual Design & Branding

- **Color Scheme:** A soft pastel palette (light blues, pinks, purples) creates a modern, friendly atmosphere.
- **Typography:** Clean, sans-serif fonts ensure readability. Headlines are bold for emphasis, while body text is easy on the eyes.
- **Imagery:** Each episode has a custom illustrated cover. All covers share a consistent art style (often a pastel gradient background with a simple icon).
- **Responsive Design:** The layout adapts to desktop, tablet, and mobile screens. On small screens, content stacks vertically and the menu collapses.
- **Dark Mode:** Users can toggle a dark theme where backgrounds turn dark gray and text turns light. The preference is saved so the theme persists.
- **Overall Aesthetic:** The interface is minimalist and content-focused, reflecting the theme of podcast production. The moody dark theme resembles an actual recording studio environment

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Conclusion

EchoHub brings together modern web design and podcast creativity. It offers an intuitive way for users to discover and enjoy audio content, while also providing tools to participate in content creation. The platform showcases our team's skills: from writing engaging scripts to coding interactive web pages. By combining multimedia, our project demonstrates how a student team can build rich, user-friendly applications using open tools and creative vision

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